Packing features — how to use them

Applies to: Warehouse Wizard 1.29.5
Audience: warehouse managers, supervisors, inventory clerks, receiving operators
Related: pallet-pack-standards-plan.md — the design and build order behind this

1. What shipped

A packaging profile used to be a carton-dimension template that nothing downstream read. It is now the **pallet build standard** for a SKU: `12 × 7` — twelve cases per layer, seven layers, 84 cases on a full pallet.

Three things follow from that one number:

- **Receiving works from it.** Selecting a SKU assigns its standard, the Qty per pallet field shows the standard beside it, and one tap fills it in.
- **The built height is real.** A pallet's height is now the deck plus every layer of cartons, not the height of one carton. Put-away and Location Moves refuse a pallet that will not fit under the bin.
- **The build is designable.** The Command Center's **Packing** tab renders the stack as you shape it, and saves the result as the SKU's standard.

Off-standard receipts are **recorded, never blocked** — a short last pallet is normal, and a blocked receipt just gets worked around invisibly.

2. The vocabulary

Term	Means	Where it comes from
Pack code	`12 × 7` — cases per layer × layers	`packages_per_layer`, `layers_per_pallet`
Cases per pallet	The 84 in `12 × 7`	Derived: per layer × layers
Units per pallet	What the receiving form counts in	Cases per pallet × units per package
Built height	Deck + every layer, slip sheets included	Derived from the carton height
Bin clearance	The bin's ceiling	The bin location's max height
Usable height	Clearance minus the safety margin	Margin defaults to **76 mm (3 in)**, set per warehouse
Conformance	Standard / short / overpack	Actual cases against the standard

Cases versus units. A pack code counts *cases*. Receiving's Qty per pallet counts *stock units*. The two coincide only when a profile's **units per package** is 1, so every screen that compares them says which unit it used. If a

SKU has no units-per-package set, the comparison is made in cases and says so.

3. Where the packing features live

Surface	Who normally uses it	What it is for
Command Center → Packing tab	Manager, supervisor	Design a build, watch the stack, save it as the SKU's standard
Receiving → Lot, batch, and packaging	Receiving operator, clerk	Record the pack code that actually arrived; create a standard on the spot
Receiving → Qty per pallet	Receiving operator	See the standard and apply it in one tap
Packaging Profiles module	Manager, admin	The full field-by-field editor; the place to *change* an existing standard
Put-Away and Location Moves	Operator	The height rule that uses all of the above

4. The Pallet Pak Designer (Command Center → Packing)

Getting in

The Packing tab sits beside Floor, Dock and Office on the Command Center. Until the feature is released it shows a padlock for everyone except developers, who see the tab with a **Developer preview** badge. See §8 for the one toggle that releases it.

Step by step

1. **Pick a SKU.** The search box at the top right of the panel matches SKU, name or barcode. If the SKU already has a pallet standard, its numbers load into the designer, so you are shaping a real build rather than starting from defaults.

2. **Shape the build with the sliders.**

Slider	Range	What it changes
Cases per layer	1–40	The first number of the pack code
Grid across	1 – cases per layer	Cases along the long side; the rest wrap
Layers	1–30	The second number of the pack code
Carton height	40–800 mm	**Drives the built height** — see the warning in §7
Actual on pallet	0 – 125% of standard	A "what if this pallet is short" preview
Bin clearance	800–3200 mm	The bin ceiling you are testing against

   Layer pattern (block, brick, pinwheel, column, custom) and **slip sheet** height sit below the sliders. The pattern is recorded on the standard and drives how the layer is drawn.

3. **Read the stage.** The pack code is printed at display size in the top left — that is the number an operator reads across a room — with the cases-per-pallet count under it. The unit switch in the top right toggles **mm / inch / ft-in** for every length on the screen.

4. **Read the two verdicts** in the right-hand rail. The first is fit:

Verdict	Means
`fits`	Clear of the usable height with room to spare
`tight`	Under 100 mm of headroom. Best-fit slotting will prefer this bin
`blocked`	Taller than the usable height. The message names the overage **and the remedy** — "Drop to 6 layers, or slot a taller bay"
`unknown`	No bin clearance to check against

   The second is conformance, driven by the **Actual on pallet** slider: `standard`, `short` — with the layer breakdown, "4 full layers plus 5 of 12 on top" — or `overpack`.

5. **Save as pack standard.** The dialog asks for four things:
   - **Profile name** — pre-filled from the pack code (`12x7`, suffixed if taken).
   - **Package type** — case, carton, box, tray or bag.
   - **Units per package** — the dialog spells the product out: "84 cases × 6 = 504 units per pallet". That is the number receiving will use.
   - **Make this the pallet standard for this SKU** — one standard per SKU. If the SKU already has one, the toggle is disabled and the panel names the profile holding it.

Two things to know before you save

- **The designer always creates a new profile.** Loading an existing standard and saving does not overwrite it — it adds a second profile alongside. To *change* an existing standard, edit it in the Packaging Profiles module.
- **Saving needs you online and permitted.** Master data is created online only, so two devices cannot race the same profile name. Without the packaging permission the Save button stays disabled with the reason underneath it.

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5. Receiving

A. The SKU already has a standard

Selecting the SKU assigns its standard automatically — no digging in a collapsed section. Beside **Qty per pallet** you get a chip reading `12 × 7 · 504`, and if the typed quantity differs, a **Use standard** link that fills it in.

A declared standard outranks the learned suggestion from prior pallets. Anything you type yourself outranks both.

B. Recording the pack that actually arrived

Open **Lot, batch, and packaging** on the line. Two fields matter:

- **Packaging** — the profile list, filtered to this line's SKU, each entry showing its pack code.
- **Pack code** — free text for what this container was built to. Accepts `12x7`, `12 X 7`, `12 × 7`, `12*7` and `12 by 7`. A bare number is rejected on purpose: `84` is a quantity, and reading it as `84 × 1` would invent a standard nobody can build.

As soon as the code parses, a conformance line appears under the section:

> `12 × 7 · short 84 units of 504 — 6 full layers plus 4 of 12 on top.`

This never blocks the receipt. It is recorded as variance, shown in amber when it is off standard.

C. Creating a standard from the floor

Create Package Standard, under the Pack code field, opens a small dialog seeded with whatever code you typed. Enter cases per layer, layers, package type, units per package and — importantly — the **carton height**. The preview beside it renders the stack, and saving assigns the new profile to the line you were working on.

The button is disabled, with the reason on hover, when there is no SKU on the line, when the device is offline, or when you do not hold the packaging permission.

D. The proposal after a receipt

When a SKU has no standard and the receipt was good evidence of one, the same dialog opens after saving, with the reason it appeared:

- *"You recorded this container as 12 × 7. Save it as the standard for this SKU?"* — a typed pack code is a declaration by someone who may set master data, and is enough on its own.
- *"5 prior pallets of this SKU came in at 504. Save 12 × 42 as the standard?"* — an observed quantity needs at least three prior pallets **and** has to divide cleanly into layers.

Declining is remembered for the rest of the session, so the prompt does not come back on the next line.

6. The Packaging Profiles module

The **Packaging** module (`/packaging-profiles`) is where an existing standard is changed. The pack fields are grouped into their own section on the form:

Field	Notes
Packages per layer / Layers per pallet	The two halves of the pack code
Layer pattern	Block, brick (alternate 90°), pinwheel, column, custom
Grid across	Cases along the long side. Blank derives a near-square grid
Package length / width / height (mm)	The carton. Height drives the built pallet height
Pallet length / width (mm)	The footprint, overhang included
Deck height (mm)	Empty pallet. Blank restores 145
Slip sheet (mm)	Tier sheet between layers. Blank restores 0
Pallet tare (kg)	Empty pallet weight, for the bin weight rule
Max stacked pallets	Whether this pallet may be double-stacked in a bin
Quantity tolerance	Units of slack before a receipt is flagged off standard
Build notes	"Labels face out", "corner posts on the top layer"
Pallet standard for this SKU	One per product. Receiving assigns it automatically

Carton dimensions typed in millimetres keep the older centimetre columns in step automatically — there is nothing to enter twice.

7. The height rule

Before this release, a pallet's height held the height of a *carton*, so a 22 cm number was compared against a 190 cm bin and every height check passed. It now holds the built height of the pallet, snapshotted at receipt.

Where it bites: Put-Away confirmation and Location Moves both refuse a pallet taller than the bin's usable height. The block is hard — there is **no override and no reason code** for a height failure, unlike temperature or mixed-SKU. The message quotes raw millimetres on both sides deliberately, because a quarter-inch display cannot show a 6 mm difference:

> Pallet is too tall for this bin — 1905 mm pallet, 1900 mm bin, 76 mm margin

Usable height = bin clearance − safety margin. The margin defaults to 76 mm (3 in) and is set per warehouse. The same arithmetic runs everywhere fit is judged, so the designer's verdict and the put-away block agree.

> **The one thing that silently disables all of this:** a standard saved with **no carton height** has no pallet height, and a pallet with no height passes every bin check. Both dialogs warn about this in amber. Fill the carton height in.

A missing height never blocks — an unrecorded measurement is not evidence of a problem — which is exactly why that warning matters.

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8. Permissions and releasing the Packing tab

Two feature codes in **Settings → Users & Roles → Role Matrix** govern this work:

Feature code	Controls
`packaging`	Creating and editing pack standards, from receiving or anywhere else
`pack_designer`	Seeing and using the Command Center Packing tab

- The matrix now **really governs** packaging profile writes, client and server. An inventory clerk granted edit can record a standard on the floor; previously the button appeared and the database refused it.
- A warehouse manager keeps write access whatever the matrix says — a missing matrix row can never lock a manager out of master data.
- **Deleting** a profile stays manager-tier. Widening who may *create* a standard was the point; widening who may destroy one was not.
- **Releasing the Packing tab** is one toggle and no deploy: set `is_released` on the `pack_designer` feature. Until then the tab is padlocked for everyone but developers, and the panel wears a *Developer preview* badge. The role grants are already seeded, so releasing is not a configuration exercise performed under pressure.

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9. Not in this release

So nobody goes looking for them:

- **The pallet label does not carry the pack code or a layer map yet.** It still prints the packaging profile name only.
- **There is no warehouse-wide fit test.** The designer checks a build against the one bin clearance you set on the slider; counting how many bins in the warehouse could actually take a standard pallet is a later phase.
- **Profiles are not versioned in the UI.** Editing a standard changes it for every pallet that references it. Where a supplier has permanently changed a pack, create a new profile rather than editing in place.
- **There is no per-receipt pack override field.** Record what arrived with the Pack code field, which is captured as variance on the receipt.
- **Slotting does not yet score on height.** Best-fit, height bands and family affinity are the next phase; today the height rule filters, it does not rank.